

Bhadra Korukonda

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SKILLS

Programming Languages: Python, Java, JavaScript

Libraries/Frameworks: PyTorch, scikit-learn, NumPy, Flask, React

Tools/Platforms: FastAPI, Flask, REST APIs, Git, Docker, Kubernetes, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, SQL

PROFESSIONAL EXPERIENCE

Research Centre Imarat(RCI), DRDO

Data Scientist Intern |

June 2024 – Aug 2024

- Worked on real missile trajectory datasets (lat/long sequences): wrote Python scripts to automatically detect and correct redundancies, fill/interpolate missing values, and clean sequences for direct use in guidance code.
- Designed and delivered a desktop Git client for an intranet environment (no public internet access), enabling scientists to perform add, commit, push and pull operations via the UI instead of using command line arguments
- This solution reduced version-control errors and onboarding time for engineers working on classified trajectory and guidance code.

IIT Hyderabad

Research Intern | Python, PyTorch Geometric, Pandas, Geopy, GRU/LSTM, GNNs (GCN, GAT, EGAT) May 2025 – Aug 2025

- Implemented a hybrid Edge-Aware Graph Attention Network (EGAT-GRU) to forecast PM2.5 levels across Delhi using spatio-temporal air quality data.
- Used wind direction and the distance between stations as edge features so the model can learn how wind carries pollutants across locations.
- Obtained competitive results (RMSE 16.72, MAE 10.05) on real-world air-quality dataset and gained experience with large-scale graph-based time-series modeling.

PROJECTS

GRAIL-LM | FastAPI, LangChain, Neo4j, Docker, Ollama (LLaMA3)

- Built a full Graph-RAG pipeline that feeds retrieved sub-graphs from Neo4j into LLaMA3 to answer multi-hop questions with proper source tracing and significantly fewer hallucinations than vanilla LLM.
- Added GNN-based retrieval using PyTorch Geometric on top of keyword and vector search so the system uses actual graph relationships and distances instead of only text similarity.
- Containerized the entire stack (FastAPI backend, Neo4j, Ollama) with docker-compose and added a Streamlit dashboard to visualize the exact reasoning path step-by-step.

Cloud Automation & Monitoring Pipeline | Docker, AWS, Prometheus, Grafana, Jenkins, Python

- Took a complete 3-tier web application (frontend, backend, database) and fully automated its entire lifecycle.
- Built a Jenkins pipeline that handles Git triggers → Docker builds → Kubernetes deployment → automated rollback on failure.
- Integrated Prometheus and Grafana dashboards for real-time monitoring of CPU, memory, latency, and deployment status.
- Configured zero-downtime deployments using Kubernetes rolling updates with readiness/liveness probes and automated rollback on failure.

Automated Resume Ranking System using NLP and MCDM | Python, Flask, spaCy, scikit-learn, PyMCDM, Tesseract OCR

- Built end-to-end resume evaluation system with Flask REST APIs, real-time processing pipeline, and interactive React dashboard.
- Implemented a unified OCR-NLP workflow using Tesseract for resume text extraction and spaCy for text cleaning, entity recognition, and feature generation, enabling accurate and automated candidate ranking.
- Implemented TOPSIS algorithm to compute weighted candidate scores, improving shortlisting efficiency and reducing human bias.

EDUCATION

Bachelor of Technology (CS Engg), Woxsen University, GPA: 3.0/4.0

Narayana Jr College(SSC), Hyderabad, Percentage: 87.1%

Chinmaya Vidyalaya(CBSE), Hyderabad Percentage: 90%

Aug 2022 – July 2026

June 2020– May 2022

May 2008 – June 2020